

AYUSH RANJAN ROY

GenAI Engineer — Multi-Agent Systems & LLM Infrastructure

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SUMMARY

GenAI engineer architecting production agent infrastructure: stateful multi-agent orchestration, MCP-based tool gateways, and cost-controlled, policy-governed routing, backed by 2 years of distributed-systems software engineering (Terraform, Kubernetes, CI/CD) at production scale. Delivered a multi-agent system from PoC to production handling 300+ documents per day at 97%+ accuracy, and built a governance workflow with rule-based enforcement and audit-traceable approvals from scratch.

EXPERIENCE

Full Stack LLM Development Analyst

Aug 2025 – Present

Accenture (GenAI & Data), Gurugram • GenAI Engineer, Multi-Agent Systems

- Co-architected a stateful, graph-based multi-agent pipeline in LangGraph orchestrating 4 LLMs (Claude Sonnet, GPT-4o, GPT-5, Azure DI) for document classification and extraction, taking it from PoC to production at 300+ docs/day, 97%+ accuracy.
- Designed an ensemble-consensus (majority-voting) classification layer with confidence-score routing through a model gateway and zero-redeployment config for live server management; lifted classification accuracy 40%, scaled batch capacity from 9 to 50 pages, and cut latency 55% (multi-page) / 30% (single-page) via async parallel inference and per-model cost control.
- Built an MCP server and tool-invocation protocol enabling autonomous LLM access to PostgreSQL and pgAdmin, extending agent capability beyond static prompting into live, governed tool use.
- Built Watchtower, an agentic-huddle UI backed by a DB queue mechanism surfacing live agent state and decision traces; engineered end-to-end IDVS integration across 9 failure scenarios (queueing, DB-fetch fallback, email drafting, reply parsing) with zero unhandled errors in production.
- Designed a from-scratch Dispute and Deduction workflow combining a rule-based policy-enforcement engine (20+ reason codes) with LangGraph human-in-the-loop checkpoints and UI integration; researched and resolved failure patterns across 20+ agents via Kubernetes (AKS) log diagnostics, OpenTelemetry, and Datadog audit trails.
- Authored architecture and onboarding documentation, ran KT and testing sessions on LangGraph/LangChain for 4 engineers, and coordinated delivery across Agile sprints, cutting ramp-up time 60%+.

Platform Engineer

Aug 2023 – Aug 2025

Accenture (Security), Gurugram • Distributed Systems & Production Infrastructure

- Drove the MxDR-to-Adaptive-MxDR platform migration as lead software engineer: Terraform IaC refactoring, GitHub Actions pipeline architecture, and live-incident production debugging across 20 microservices serving 800+ enterprise customer accounts; cut deployment turnaround 35% and infra cost 30%, directing a 4-member team.
- Owned Terraform IaC and server management across 6 environments, modernized 8+ AWS Lambda functions (Python 3.8 to 3.12) cutting runtime failures 50%, and deployed a VPN-secured OpenTelemetry + BindPlane observability protocol with 5-min synthetic gRPC health monitoring for centralized SIEM audit trails.

PROJECTS

IncidentIQ — AI-SRE Multi-Agent Platform [\[GitHub\]](#) — *LangGraph, FastAPI, pgvector, RBAC*

- Manual incident triage is slow and error-prone under pressure across noisy, correlated alerts.
- Built a 5-agent LangGraph pipeline (triage, context, RAG knowledge, root-cause synthesis, supervision) producing confidence-scored diagnoses.
- Validated end-to-end with full audit logging: alert ingestion, diagnosis, human-gated rollback approval, and execution.

RAGGym — RAG Interview-Prep Platform [\[Live Demo\]](#) — *LangGraph, Qdrant, RAGAS, Streamlit*

- Most RAG systems lock into one vendor stack and answer confidently even on weak retrievals.
- Built a fully swappable architecture (LLM, embeddings, vector store hot-configurable) with a self-correcting retrieval loop that grades and re-queries before answering.
- Added an AI-graded practice mode turning passive reading into hands-on exercises; deployed live with a RAGAS evaluation harness.

TECHNICAL SKILLS

Agent & Orchestration: LangGraph, LangChain, Multi-Agent Orchestration, MCP, Human-in-the-Loop, Stateful Workflow Design

LLMOps & Governance: Model Gateway/Routing, Cost Control, Policy Enforcement, Audit Trails, OpenTelemetry, Datadog, CloudWatch

Models & Retrieval: Claude Sonnet, GPT-4o/5, Azure AI Foundry, RAG, pgvector, PostgreSQL

Infrastructure: Python (async), FastAPI, Docker, Terraform, AWS (Lambda, EC2, S3), Azure AKS, GitHub Actions, CI/CD

CERTIFICATIONS

Claude Certified Architect – Foundations

Anthropic, May 2026

EDUCATION

B.Tech, Electronics and Communication Engineering

Jul 2019 – Aug 2023

Vellore Institute of Technology, Tamil Nadu, India